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SHENZHEN SILICONTRA TECHNOLOGY CO.,LTD.



PAN3029ATR4-GC

433MHz Wireless Transceiver Module User Specification

(V1.0)

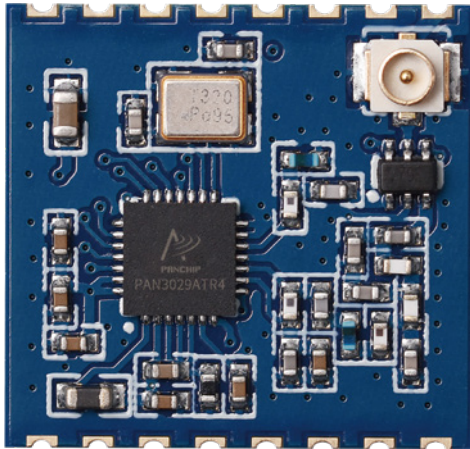
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Document Revision History

Version	Change date	Change Notes
V1.0	October 23, 2024	Initial version

1 Module Introduction



(Front of the module)



(Back of the module)

(The actual module shall prevail)

1.1 Module Overview

The PAN3029ATR4-GC is a module independently developed and designed by our company based on the PAN3029 RF chip from the domestic manufacturer, Pantech Microelectronics.

The biggest highlight of the PAN3029ATR4-GC module is its transmit power reaching +20dBm and receive sensitivity reaching -143dBm. Combined with its advanced ChirpLoTTM modulation and demodulation technology, it significantly improves communication range. It holds a strong position in short-range IoT wireless communication applications.

The PAN3029ATR4-GC module features small size, low power consumption, long transmission distance, and strong anti-interference capabilities, making it widely applicable in various wireless communication fields within the IoT ecosystem.

1.2 Module Features

- Supports 433MHz band with strong penetration
- Maximum transmit power 20dBm, software-adjustable receive current 4.1mA@DCDC mode
- SPI communication interface, directly connectable to various microcontrollers, wide operating voltage range 1.8~3.6V
- Industrial-grade standard design, supports long-term use at -40~+85°C, ultra-small size, only 16x17mm
- Stamp-perforation design for easy mass production

1.3 Application Scenarios

- 家庭安防报警
 - 智能家居
 - 智能停车系统
 - 农业自动化解决方案
 - 无线工业级遥控器，工业传感器等产品
- 医疗保健
 - 楼宇自动集抄系统
 - 汽车行业应用

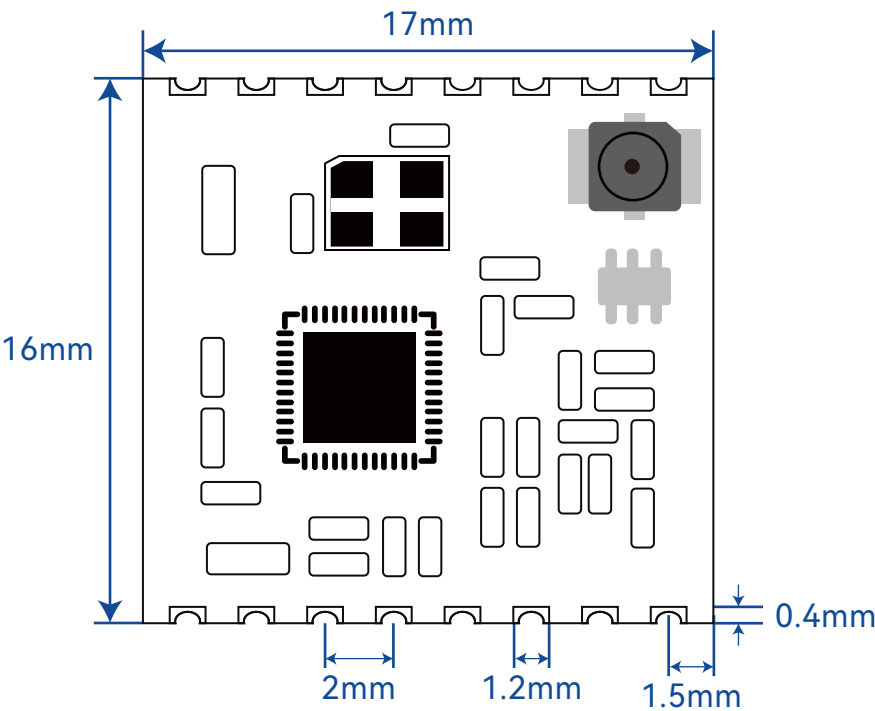
2 Module Parameters

2.1 Basic Electrical Parameter Diagram of the Module

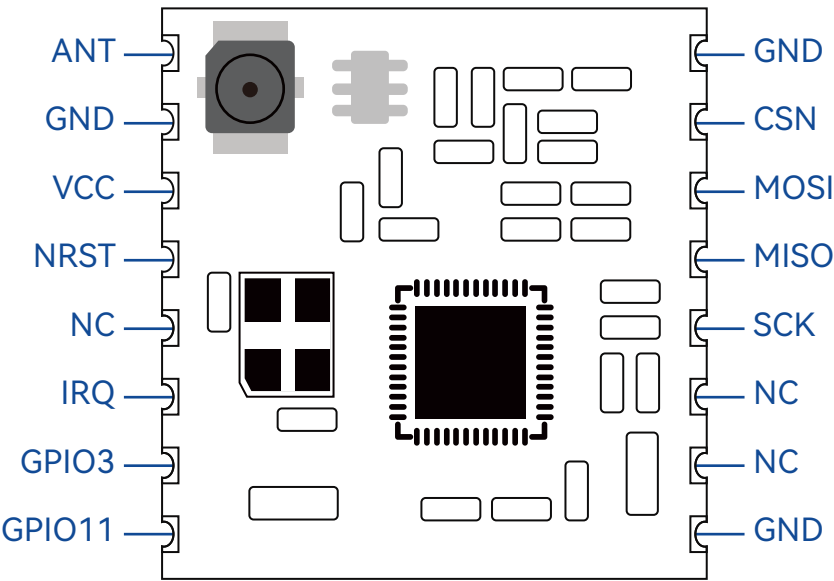
项目	规格		说明
参数	性能		备注
工作电压	1.8 ~ 3.6V		标准 3.3V
工作温度	-40 ~ 85℃		
工作频率	408 ~ 550MHz		推荐 433MHz，可在程序中配置
功耗	发射状态	96mA@20dBm	最大发射功率
	接收状态	8mA@LDO / 4.1mA@DC-DC	
	睡眠状态	260nA	
发射功率	20dBm		最大值，用户可编程调节
接收灵敏度	-143dBm		@62.5KHz
调制方式	ChirploT™		
支持	BW: 62.5KHZ~500KHZ; SF: 5~12; CR: 4/5, 4/6, 4/7, 4/8		
接口方式	邮票孔		
通讯接口	SPI		
外形尺寸	16*17mm		
天线类型	邮票孔外置天线、IPEX座子		等效阻抗约 50Ω

3 Module Description

3.1 Module Dimension Diagram



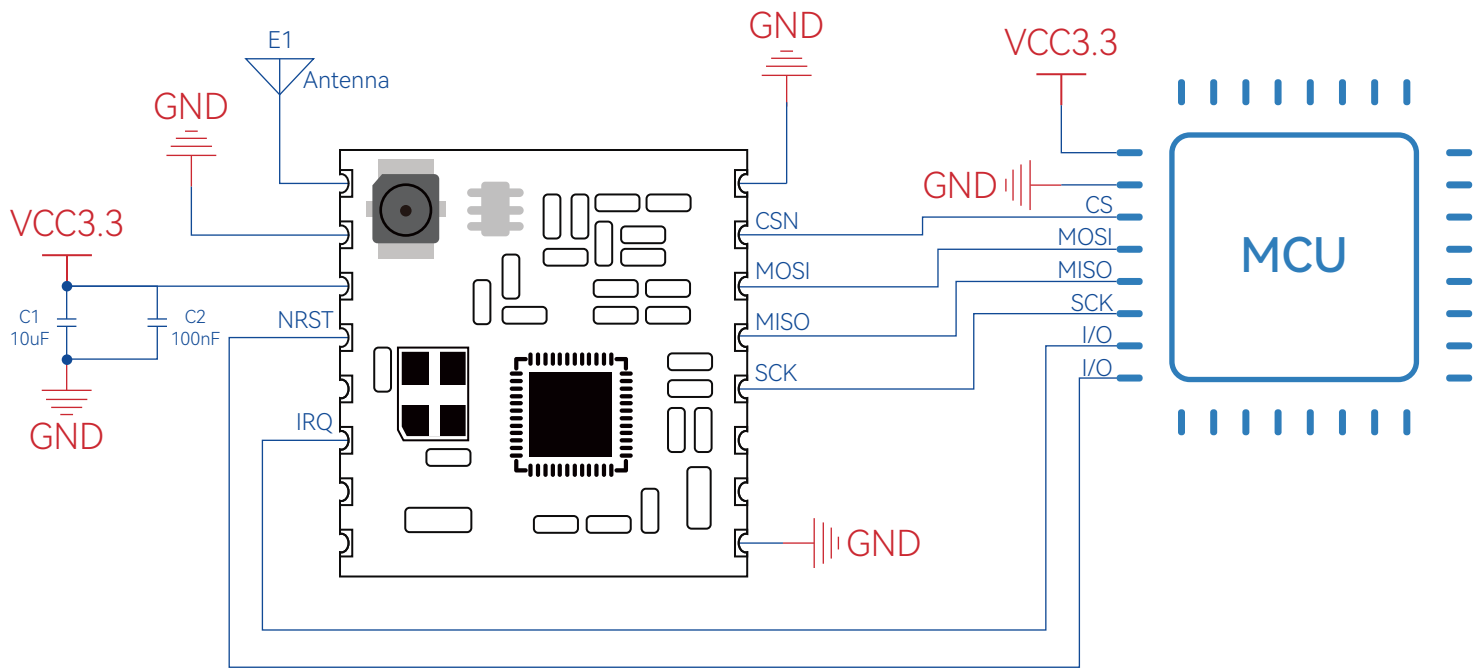
3.2 Module Pin Function Definition Diagram



3.3 Pin Function Description

序号	接口名	功能描述
1	ANT	天线接口，等效阻抗约50Ω
2	GND	地
3	VCC	模块电源引脚，标准供电电压3.3V
4	NRST	复位引脚 低电平有效
5	NC	内部悬空脚
6	IRQ	中断信号
7	GPIO3	数字IO
8	GPIO11	数字IO或者信道状态指示信号
9	GND	地
10	NC	内部悬空脚
11	NC	内部悬空脚
12	SCK	SPI总线时钟引脚, 常态为高电平
13	MISO	SPI总线从机输出引脚。当 MCU 读取模块数据时, 数据由此引脚输出
14	MOSI	SPI总线从机输入引脚。当 MCU 向模块写数据时, 数据从此引脚写入
15	CSN	SPI片选引脚, 低电平有效
16	GND	地

3.4 Module Connection Diagram



4 Additional Notes

- 1、推荐使用线性稳压电源(LDO)对该模块进行供电，电源纹波系数尽量小，模块需可靠接地，并注意电源正负极的正确连接，如反接可能会导致模块永久性损坏；
- 2、模块天线附近不能围绕其它金属物体，否则会严重影响通讯距离。

5 Antenna Selection

5.1 Antenna Usage Precautions

- 天线安装结构对模块性能有较大影响，需要更好的效果需要天线外露，最好垂直向上。当模块安装于机壳内部时，可使用优质的天线延长线，将天线延伸至机壳外部;如遇产品不允许外露就需要匹配弹簧天线或者FPC天线。
- 天线如安装在金属壳内部，将导致传输距离极大削弱。
- 如选购吸盘天线，引线尽可能拉直，吸盘底盘尽可能吸附在金属物体上。



433MHz弹簧天线



433MHzFPC天线



433MHz棒状天线



433MHz吸盘天线

6 Hardware Design

- 推荐使用直流稳压电源对该模块进行供电，电源纹波系数尽量小，模块需可靠接地。
- 请注意电源正负极的正确连接，如反接可能会导致模块永久性损坏。
- 请检查供电电源，确保在推荐供电电压之间，如超过最大值会造成模块永久性损坏。
- 请检查电源稳定性，电压不能大幅频繁波动。
- 在针对模块设计供电电路时，往往推荐保留30%以上余量，有利于整机长期稳定地工作。
- 模块应尽量远离电源、变压器、高频走线等电磁干扰较大的部分。
- 高频数字走线、高频模拟走线、电源走线必须避开模块下方，若实在不得已需要经过模块下方，假设模块焊接在 Top Layer，在模块接触部分的Top Layer铺地铜（全部铺铜并良好接地），必须靠近模块数字部分并走线在 Bottom Layer。
- 假设模块焊接或放置在 Top Layer，在 Bottom Layer 或者其他层随意走线也是错误的，会在不同程度影响模块的杂散以及接收灵敏度。
- 假设模块周围有存在较大电磁干扰的器件也会极大影响模块的性能，跟据干扰的强度建议适当远离模块，若情况允许可以做适当的隔离与屏蔽。
- 假设模块周围有存在较大电磁干扰的走线（高频数字、高频模拟、电源走线）也会极大影响模块的性能，跟据干扰的强度建议适当远离模块，若情况允许可以做适当的隔离与屏蔽。

7 Unsatisfactory Transmission Distance

- Communication distance will decrease when there are obstacles or obstructions in line-of-sight communication.
- Temperature, humidity, and co-channel interference can increase packet loss rate.
- Ground absorption and reflection of radio waves result in poor performance when tested near the ground.
- The presence of metal objects near the antenna, or placement within a metal enclosure, will cause severe signal attenuation. An excessively high airspeed setting (higher airspeed results in shorter distances) can also cause problems.
- Lower-than-recommended power supply voltage at room temperature reduces transmission power.
- Poor antenna-module compatibility or antenna quality issues.

8 Module Easily Damaged

- Please check the power supply and ensure it is within the recommended voltage range. Exceeding the maximum value will cause permanent damage to the module.
- Please check the power supply stability; the voltage should not fluctuate.
- Please ensure anti-static operation during installation and use, as high-frequency components are electrostatically sensitive.
- Please ensure that the humidity is not too high during installation and use, as some components are humidity-sensitive.
- Unless there are special requirements, it is not recommended to use the product at excessively high or low temperatures.

9 High Bit Error Rate

- Interference from nearby co-channel signals: Move away from the interference source or change the frequency or channel to avoid interference.
- Inadequate power supply can also cause garbled characters; ensure power supply reliability.
- Poor quality or excessively long extension cables or feeders can also lead to a higher bit error rate.